Texas A&M University-Corpus Christi

College of Science

BIOL 4360 - COMPUTATION FOR 21ST CENTURY B

Department of Life Sciences

Fall Full Term 2026

[FALL-26] BIOL-4360-001 - COMPUTATION FOR 21ST CENTURY B

Course Information

Meeting Time: F 02:00pm - 04:30pm
Meeting Location: CCH 206
Instructional Method: Face-to-Face/Web Enhanced 25%>
Course Website: https://canvas.tamucc.edu/

This course uses hands-on computing activities and requires regular access to a laptop or desktop computer suitable for content creation and computational work. A Windows, Mac, or Linux computer is required. Chromebooks, iOS devices, and Android devices are not appropriate as the primary computing device for this course.

Students will use open-source computing tools and may use platforms such as Canvas, GitHub, R/RStudio, Python, command-line Linux tools, and other instructor-approved software. Students should expect to install software, manage files, use online repositories, submit assignments electronically, and work with biological datasets.

No online proctoring is required unless announced by the instructor.

Instructor Information

Christopher Bird

Email: chris.bird@tamucc.edu
Office Phone Number: 361-825-6024
Student Hours: M: 4-5, TW: 4-5:30
Office Location: TH234

Appointments: A student may make an appointment to see me at times other than the scheduled office hours. I am available for consultation and extra help, but it is the student's responsibility to request such help.

Course Description

This course is designed to prepare and enable students to use computational tools for bioinformatic applications in advanced courses and independent research projects. Students will be introduced to powerful open-source computing tools used in biological research for creation, organization, manipulation, processing, analysis, and archiving of big data. While not a formal requirement, it is assumed that students have a firm command of basic algebra. Juniors or Seniors only. Offered every Fall. Stacked with BIOL 5360.

Course Overview

This 3-credit course introduces students to open-source computational tools used in biological research for the creation, organization, manipulation, processing, analysis, visualization, and archiving of biological data. The course is designed to prepare students to use computational tools for bioinformatic applications in advanced courses, independent research, and data-intensive biological projects.

Major topics include biological data repositories and data formats, command-line Linux computing, Bash scripting, regular expressions, supercomputing, data wrangling with Bash and R, data visualization with R and Python, version control with Git and GitHub, scientific typesetting with Markdown and/or LaTeX, and the use of large language models as tools to assist with computational workflows.

The course combines short lectures with interactive, hands-on computing exercises. Students will learn by doing: installing tools, writing commands and scripts, manipulating datasets, producing figures, managing projects, and completing weekly assignments that reinforce computational skills.

Prerequisites and Corequisites

There are no prerequisites for this course.

Required Materials

No paid textbook is required.

Additional Information

Students are required to have access to the following:

- **R for Data Science**, by Hadley Wickham, Mine Çetinkaya-Rundel, and Garrett Grolemund. Free online version: https://r4ds.hadley.nz/
- A Windows, Mac, or Linux computer suitable for computational work.
- Recommended minimum computer specifications: at least 8 GB RAM and 512 GB storage.
- Reliable internet access.
- TAMU-CC email and Canvas access.
- Access to the Fall 2026 course GitHub repository or other course website provided by the instructor.
- Required open-source software announced by the instructor, which may include Git, GitHub, R, RStudio, Python/Miniconda, command-line tools, and related packages.

Students who do not have access to an appropriate computer should contact Dr. Bird as early as possible so that available resources can be discussed.

Recommended Materials

The following materials are recommended, but not required

Additional Information

- **Computing Skills for Biologists: A Toolbox**, by Stefano Allesina and Madlen Wilmes.
- Course resources provided through the course GitHub repository.
- Online documentation for Linux/Unix commands, Git, GitHub, R, tidyverse, Python, Biopython, Markdown, and LaTeX.
- A system for backing up coursework, such as OneDrive, GitHub, or another reliable backup method.
- A plain-text editor or integrated development environment suitable for coding.

Student Learning Outcomes (SLO)

By the end of this course, students will be able to:

1. Recognize, describe, and organize data into tidy data structures.
2. Locate scientific data repositories and download data.
3. Operate UNIX/Linux computers from the command line.
4. Construct and modify computer programming and scripting logic structures for processing biological data.
5. Describe and use regular expressions to query data.
6. Use version control software, including Git and GitHub, to organize and manage projects.
7. Typeset scientific or technical documents with LaTeX and/or Markdown.
8. Use common open-source tools for biological data manipulation, including Bash, R/tidyverse, and Python.
9. Use large language models effectively and responsibly to assist with data processing, troubleshooting, and computational workflows.

Instructional Methods & Activities

Classroom time will involve a mix of lecture, demonstration, discussion, and interactive hands-on computing exercises. Students will receive course information through lectures, live coding demonstrations, guided practice, course website materials, readings, software documentation, and weekly assignments.

Activities may include command-line exercises, scripting, data wrangling, data visualization, use of biological data repositories, Git/GitHub workflows, supercomputing demonstrations, R and Python exercises, scientific typesetting, and applied problem-solving using biological datasets.

Weekly assignments are designed to reinforce concepts introduced in class and help students build practical computational skills over time.

Major Course Requirements

Participation	15%
Assignments	40%
Exam 1	15%
Exam 2	15%
Final Exam	15%
Total	**100%**

Course Content/Schedule

Date	Lecture Topic	HW Due
	Theme I: Welcome to the Matrix	
Wk 0	1. Course overview 2. Biological Data Repositories, Structures, Formats 3. Computer set up	
Wk 1	Linux Boot Camp I 1. UNIX philosophy 2. Navigating/creating/ manipulating directories & files 3. How to get help: man pages 4. Basic commands: cd, ls, cp, mv, mkdir, rm, tr, cut, cat, head, tail, 5. Commands useful for manipulating data files in text streams	Assignment 0

Wk 2	Linux Boot Camp II 1. Wildcards, substituting characters, permissions, sudo 2. Pattern matching with grep & regex 3. Intro to Computer Programming: Shebang!, Scripting, For Loops	Assignment 1
Wk 3	Linux Boot Camp III 1. More Computer Programming with bash 2. Logic: if-then-else; looping with while, and GNU parallel 3. Functions: diy commands 4. Advanced text stream manipulation: sed, paste, …	Assignment 2 SuperComputer Acct
Wk 4	Version Control & Supercomputing 1. Linux repositories and tools for biologists 2. Version control with git 3. Super computing, ssh	Assignment 3
	Theme II: Wrangling and Visualizing Data With R	

Wk 5	R Boot Camp I 1. R Philosophy 2. Command line R 3. R data types & structures 4. Math, equalities, logic 5. Basic statistical functions 6. Reading and writing data	**Exam 1** Install R & R Studio
Wk 6	R Boot Camp II 1. Scripting & writing good code 2. Loops & if-then decision logic 3. R Studio 4. Functions 5. Libraries 6. Random numbers 7. Vectorized loops 8. Debugging 9. More basic stats 10. Base R plots	Assignment 5
Wk 7	R Boot Camp III 1. tidyverse 2. Basic reading & manipulating data 3. Basic computing statistics 4. Visualization of data w ggplot2	Assignment 6
Wk 8	R Boot Camp IV 1. Advanced tidyverse, pipelines 2. Manipulating & wrangling data	Assignment 7
	Theme III: Programming the Matrix	

Wk 9	Python Boot Camp I 1. Intro to Python 2. Data structures 3. Functions 4. Decision logic and loops 5. Reading and writing files	**Exam 2** Install Miniconda
Wk 10	Python Boot Camp II 1. Writing code 2. Modules & Program Structure 3. Errors and exceptions 4. Debugging 5. Testing & Profiling	Assignment 9
Wk 11	Scientific Computing w/ Python 1. NumPy and SciPy 2. Pandas 3. Biopython 4. Other modules	Assignment 10
Wk 12	Scientific Typesetting w/ Latex 1. Latex document structure 2. Typsetting 3. Latex packages for biologists	Assignment 11
Wk 13	Putting It All Together	Assignment 12
Last Day of Classes (Dec 3)		Assignment 13
Final	**Final Exam: Becoming THE ONE**	

University Policies

Disabilities Accommodations

The Americans with Disabilities Act (ADA) is a federal anti-discrimination statute that provides comprehensive civil rights protection for persons with disabilities. Among other things, this legislation requires that all students with disabilities be guaranteed a learning environment that provides for reasonable accommodation of their disabilities. If you believe you have a disability requiring an accommodation, please call or visit Disability Services at (361) 825-5816 in Corpus Christi Hall 116.

If you are a returning veteran and are experiencing cognitive and/or physical access issues in the classroom or on campus, please contact the Disability Services office for assistance at (361) 825-5816.

Student Grade Appeals

As stated in university procedure 13.02.99.C0.03, Student Grade Appeals, a student who believes that their final grade reflects academic evaluation which is arbitrary, prejudiced, or inappropriate in view of the standards and practices outlined in the class syllabus, may appeal the final grade given for the course. The burden of proof is upon the student to demonstrate the appropriateness of the appeal. Student are encouraged to first discuss their final grade with the instructor prior to filing a grade appeal. For complete details on the grade appeal process, including the timeline, please review university procedure 13.02.99.C0.03, Student Grade Appeals. For assistance and/or guidance in the grade appeal process, students may contact the Dean's/Director's office in the college/school in which the course is taught.

Campus Emergencies

At TAMU-CC, your safety is a top concern. We actively prepare for natural disasters or human-caused incidents with the ultimate goal of maintaining a safe and secure campus.

- For any emergency, dial the University Police Department (UPD) at **361-825-4444** or dial 911. It's a good idea to have the UPD emergency number (and non-emergency number 361-825-4242) saved in your cell phone.
- There are nearly 200 classroom telephones throughout campus. If you feel threatened or need help and don't have a cell phone, dial 4444 (emergency) or 4242 (non-emergency) to be connected to UPD.
- If we hear a fire alarm, we will immediately evacuate the building and proceed outside and away from the building a minimum of one hundred feet.

    - Proceed to the nearest building exit or evacuation stairway. Do not use the elevator. Persons who need help navigating stairs should proceed to a marked Area of Rescue Assistance, if possible.
    - Persons with disabilities should speak with their faculty about how to best assist them in case of an emergency.
    - Review the evacuation route (see posted building evacuation map).
- TAMU-CC employs the Code Blue Emergency Notification System, an alert system which connects the campus community during emergency situations.
    - The notifications include emails, text and pre-recorded messages, as appropriate.
    - Code Blue emergencies may include severe weather warnings, threats, school closures, delays, evacuations and other incidents which disrupt regular campus activities.
    - Students can update personal contact information anytime at the Code Blue notification website.
- Shelter in Place via Code Blue.
    - "Shelter-in-place" means to take immediate shelter where you are and may be implemented for severe weather, hazardous material spills, active shooters or other dangerous situations.
    - If there is a shelter in place for a **tornado warning**, our preferred location is the bottom floor of this building, away from windows and doors.
- Active Threat Protocol. There are three things you could do that make a difference if there is an active threat: Run, Hide, and/or Fight. For more information about the Run, Hide, Fight protocol, including what to do when law enforcement arrives, visit **page 3** of the Quick Reference Guide to Campus Emergencies.

For the *Quick Campus Guide to Campus Emergencies* (including a list of Areas of Rescue Assistance and additional protocols on assisting persons with physical disabilities, hurricanes, bomb threats, animal bites, crime reporting, elevator entrapment, etc.), visit Quick Reference Guide to Campus Emergencies.

Civil Rights Reporting

Texas A&M University-Corpus Christi is committed to fostering a culture of caring and respect that is free from discrimination, relationship violence and sexual misconduct, and ensuring that all affected students have access to services. For information on reporting Civil Rights complaints, options and support resources (including pregnancy support accommodations) or university policies and procedures, please contact the University Title IX Coordinator, ext. 5826, or visit the Compliance Services Website

Limits to confidentiality. Essays, journals, and other materials submitted for this class are generally considered confidential pursuant to the University's student record policies. However, students should be aware that University employees, including instructors, are not able to maintain confidentiality when it conflicts with their responsibility to report alleged or suspected civil rights discrimination that is observed by or made known to an employee in the course and scope of their employment. As the instructor, I must report allegations of civil rights discrimination, including sexual assault, relationship violence, stalking, or sexual harassment to the Title IX Coordinator if you share it with me. These reports will trigger contact with you from the Civil Rights/Title IX Compliance office who will inform you of your options and resources regarding the incident that you have shared. If you would like to talk about these incidents in a confidential setting, you are encouraged to make an appointment with counselors in the University Counseling Center.

College Policies

Statement of Academic Continuity

In the event of an unforeseen adverse event, such as a major hurricane and classes could not be held on the campus of Texas A&M University-Corpus Christi; this course would continue through the use of Canvas and/or email. In addition, the syllabus and class activities may be modified to allow continuation of the course. Ideally, University facilities (i.e., emails, web sites, and Canvas) will be operational within two days of the closing of the physical campus. However, students need to make certain that the course instructor has a primary and a secondary means of contacting each student.

Academic Integrity/Plagiarism

University students are expected to conduct themselves in accordance with the highest standards of academic honesty. Academic misconduct for which a student is subject to penalty includes all forms of cheating, such as illicit possession of examinations or examination materials, falsification, forgery, complicity, or plagiarism. (Plagiarism is the presentation of the work of another as one's own work.) The instructor of record determines the penalty for academic misconduct or complicity in an act of academic misconduct on an assignment or test.

Classroom/Professional Behavior

Texas A&M University-Corpus Christi, as an academic community, requires that each individual respect the needs of others to study and learn in a peaceful atmosphere. Under Article III of the Student Code of Conduct, classroom behavior that interferes with either (a) the instructor's ability to conduct the class or (b) the ability of other students to profit from the instructional program may be considered a breach of the peace and is subject to disciplinary sanction outlined in article VII of the Student Code of Conduct. Students engaging in unacceptable behavior may be instructed to leave the classroom. This prohibition applies to all instructional forums, including classrooms, electronic classrooms, labs, discussion groups, field trips, etc.

Statement of Civility

Texas A&M University-Corpus Christi has a diverse student population that represents the population of the state. Our goal is to provide you with a high-quality educational experience that is free from repression. You are responsible for following the rules of the University, city, state and federal government. We expect that you will behave in a manner that is dignified, respectful and courteous to all people, regardless of sex, ethnic/racial origin, religious background, sexual orientation or disability. Behaviors that infringe on the rights of another individual will not be tolerated.

Mental Health and Well-Being Services

The university aims to provide students with essential knowledge and tools to understand and support mental health. As part of our commitment to your well-being, we offer access to Telus Health, a service available 24/7/365 via chat, phone, or webinar. Scan the QR code to download the app and explore the resources available to you for guidance and support whenever you need it.

Dropping a Class

We hope that you never find it necessary to drop a class. However, events can sometimes occur that make dropping a course necessary or wise. Please consult with your academic advisor, the Financial Aid Office, and instructor, before you decide to drop a course.

Just stopping attendance and participation WILL NOT automatically result in your being dropped from the class. Should dropping the course be the best course of action, you must initiate the process by following the instructions on the Registrar's Dropping/ Withdrawing page. Please check the Academic Calendar for the last date to drop a class.

Academic Advising

The College of Science requires that students meet with an Academic Advisor as soon as they are ready to declare a major. The Academic Advisor will set up a degree plan, which must be signed by the student, a faculty mentor, and the department chair. The Islander Advising Center is located in the Faculty Center 104 or can be reached at (361) 825-3453 or use the Academic Advising email.

Artificial Intelligence Statement - AI

AI Allowed with Disclosure – Students may use Artificial Intelligence tools to support their coursework, but all uses must be properly cited. For example, if AI was used to generate text, brainstorm ideas, or summarize content, the student must acknowledge this in the assignment. Failure to provide disclosure will be treated as a violation of academic integrity.

Course Policies

Attendance and Tardiness

Attendance is expected. Students are responsible for all material covered in class, including lectures, demonstrations, discussions, hands-on exercises, announcements, assignment instructions, and schedule changes. If you are late, enter without disrupting class and take responsibility for catching up. If you cannot attend a class meeting, make arrangements with Dr. Bird.

Participation

Participation is required and includes attendance, engagement in hands-on computing exercises, preparation for class, responsiveness during class activities, and completion of in-class work or quizzes. Because this course emphasizes practical skills, active participation is essential.

Late Work
Late assignments will lose 10% of the total possible score per day late unless other arrangements are approved by the instructor.

Make-Up Exams
Students should inform the instructor as soon as they know they will miss an exam. Make-up exams may be arranged with the instructor when appropriate.

Extra Credit
There is no separate extra credit policy. Extra points may be built into exams or assignments at the instructor's discretion.

Laptop / Computer Use
A laptop or appropriate computer is required for class. Students should bring a working computer to each class meeting unless instructed otherwise.

Cell Phone Use
Cell phones may be required for some course-related activities. Otherwise, phone use should not interfere with participation, attention, or classroom activities.

Food in Class
Food is not allowed in the computer lab.

Communication
Students should use TAMU-CC email or other instructor-approved communication platforms for course communication. Students are responsible for checking course announcements, email, Canvas, and the course website regularly.

Artificial Intelligence Tools
Students are expected to use artificial intelligence tools, including large language models, in this course. Students are responsible for the content they submit, not the AI tool. If AI is used in a novel or substantial way to complete a task, students should describe how it was used so that the instructor can evaluate the work appropriately and incorporate effective uses into future lectures or assignments.

Course Changes
The instructor reserves the right to modify course information, schedule, assignments, deadlines, and course policies when necessary. Changes will be announced in a timely manner.

IT Support Information

IT Support Information
361-825-2692 or 866-353-2491
Serviceportal.tamucc.edu

Campus Resources are provided within Canvas

General Disclaimer

I reserve the right to modify the information, schedule, assignments, deadlines, and course policies in this syllabus if and when necessary. I will announce such changes in a timely manner during regularly scheduled lecture periods.